

Integrals – 3

Differentiation under the integral sign and Frullani integrals

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Leibniz rule). If $f(x, \alpha)$ and $\partial_\alpha f$ are continuous on $[a, b] \times [\alpha_1, \alpha_2]$, then $\frac{d}{d\alpha} \int_a^b f(x, \alpha) dx = \int_a^b \partial_\alpha f(x, \alpha) dx$. For an improper integral it suffices that $\int \partial_\alpha f$ converges uniformly in α .

Theorem 2 (Frullani). If f is continuous on $[0, \infty)$ and both $f(0)$ and $f(\infty) = \lim_{x \rightarrow \infty} f(x)$ exist, then

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

Fact 1. The working scheme: introduce a parameter so that for one of its values the integral is known and the derivative in the parameter is elementary; then integrate back, remembering the constant of integration, which is recovered from a limiting value.

1. Prove Frullani's theorem and compute $\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx$, $\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx$ and $\int_0^1 \frac{x^b - x^a}{\ln x} dx$ (the last one by differentiating in the parameter).
2. Compute $I(\alpha) = \int_0^\infty e^{-x^2} \cos(\alpha x) dx$ by deriving the differential equation $I' = -\frac{\alpha}{2} I$; obtain $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ as a special case.
3. Compute $\int_0^\pi \ln(1 - 2a \cos x + a^2) dx$ for all real a by differentiating in a ; treat the cases $|a| < 1$, $|a| = 1$ and $|a| > 1$ separately.
4. Compute $\int_0^\infty \frac{\sin x}{x} dx$ by inserting the factor $e^{-\alpha x}$ and letting $\alpha \rightarrow 0^+$; justify the passage to the limit carefully.
5. (Integration Bee) Gaussian integrals and their relatives:

- a) $\int_{-\infty}^{\infty} e^{-2x^2-5x-3} dx$ (2017);
- b) $\int_{-\infty}^{\infty} e^{-(x-2024)^2/4} dx$ (2024);
- c) $\int_0^{\infty} x^3 e^{-x^2} dx$ (2016) and $\int_0^{\infty} x^5 e^{-x^4} dx$ (2020);
- d) $\int_0^{\infty} e^{-2019/(4t^2)} \frac{dt}{t^2}$ (2019).